

ORF307: Optimization

Precept: #6

Agenda

- LP reformulations
- Portfolio least squares

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Formulate the following problem as an LP

$$\min_x \sum_{i=1}^m \max\{0, a_i^T x + b_i\} \quad (1)$$

Solution:

Define the new auxiliary variables:

$$y_i = \max\{0, a_i^T x + b_i\}, \quad \text{for } i = 1, 2, \dots, m$$

Then optimization problem (1) can be written as the following constrained optimization problem

$$\begin{aligned} \min \quad & \sum_{i=1}^m y_i \\ \text{s.t.} \quad & y_i = \max\{0, a_i^T x + b_i\}, \quad i = 1, 2, \dots, m \end{aligned} \quad (2)$$

We can convert the constraint $y_i = \max\{0, a_i^T x + b_i\}$ to two inequality constraints: $y_i \geq 0$ and $y_i \geq a_i^T x + b_i$. Hence the constrained optimization problem (2) can be written as an LP problem

$$\begin{aligned} \min \quad & \mathbf{1}^T y \\ \text{s.t.} \quad & a_i^T x - y_i \leq -b_i, \quad i = 1, 2, \dots, m \\ & y \geq 0 \end{aligned} \quad (3)$$

Remark: Always remember that you can remove $\max\{\cdot\}$, $|\cdot|$ and $\|\cdot\|_\infty$ by introducing auxiliary variables and extra affine inequality constraints.

Remark: The function $r(x) = \max\{0, x\}$ is known as the Rectified Linear Unit (ReLU) function. This function plays a fundamental role in modern deep learning and neural networks, where it serves as the most commonly used activation function.

Portfolio optimization problem

- Current portfolio: $h^{curr} \in \mathbb{R}^n$ is the amount invested in n assets
- Current total value: $\mathbf{1}^T h^{curr}$
- $h - h^{curr}$ is the trade vector
- The n assets are divided into m industry sectors (e.g., financial, pharmaceuticals, semiconductors, etc.)
- $s \in \mathbb{R}^m$ denotes the dollar value sector exposure. We can write

$$s = Sh, \quad \text{where } S \in \mathbb{R}^{m \times n} \quad \text{and} \quad S_{ij} = \begin{cases} 1 & \text{if asst } j \text{ is in sector } i \\ 0 & \text{otherwise} \end{cases}$$

- The new portfolio must have a given (desired) sector exposure described by the vector $s^{des} \in \mathbb{R}^m$
- **Task #1:** minimize the trading cost given by:

$$\sum_{i=1}^n \kappa_i (h_i - h_i^{curr})^2$$

where $\kappa_i \geq 0$, $i = 1, \dots, n$ are some weights related to the trading cost of asset i .

- **Task #2:** Explain how to find h using constrained least squares.

Solution:

The optimization problem we need to solve is defined as follows

$$\begin{aligned} \min_h \quad & \sum_{i=1}^n \kappa_i (h_i - h_i^{curr})^2 \\ \text{s.t.} \quad & Sh = s^{des} \\ & \mathbf{1}^T h = \mathbf{1}^T h^{curr} \end{aligned}$$

This problem can be formulated as the following constrained linear least squares problem

$$\begin{aligned} \min_h \quad & \|Kh - Kh^{curr}\|_2^2 \\ \text{s.t.} \quad & \begin{bmatrix} S \\ \mathbf{1}^T \end{bmatrix} h = \begin{pmatrix} s^{des} \\ \mathbf{1}^T h^{curr} \end{pmatrix} \end{aligned} \tag{4}$$

where the matrix $K \in \mathbb{R}^n$ is the diagonal matrix

$$K = \text{diag}\{\sqrt{\kappa_1}, \dots, \sqrt{\kappa_n}\}$$

This can be easily derived by noting that the objective can be rewritten as follows:

$$\sum_{i=1}^n \kappa_i (h_i - h_i^{curr})^2 = \sum_{i=1}^n (\sqrt{\kappa_i} h_i - \sqrt{\kappa_i} h_i^{curr})^2$$

and performing the algebraic computations.

Exercise: Write the constrained least square in the standard form and define the matrices and vectors you used.

Remark: The objective function in this problem is designed to achieve the portfolio's financial goals (h meeting the sector and budget constraints) while minimizing the transaction costs.

- By minimizing this objective function, the trader will compute the closest new portfolio h to the current portfolio h^{curr} , subject to meeting all the required constraints.
- The objective is the sum of squared functions. This means that the cost of trading increases disproportionately as the size of the trade increases. This penalizes large, sudden trades (which can incur higher market impact costs) and favors smaller, smoother rebalancing actions.
- The weights κ_i allow us to assign different penalties to the trading of certain assets compared to others. For example, an asset that is not traded in high volumes will have a higher κ_i than an asset that has higher liquidity.

Penalize shorting assets

Suppose you want to include a penalty on shorting assets. How might you include this as a linear constraint?

Solution:

If we are allowed to short asset i then $h_i < 0$. In that case the penalty per unit of asset i is $v_i > 0$. Then we pay $v_i |h_i|$ dollars. Of course if we go long (i.e., do not short) asset i , then we do not pay any penalty.

Hence the penalty we pay can be expressed as

$$\max\{-v_i h_i, 0\}$$

Then the constraint linear least squares problem becomes

$$\begin{aligned} \min_h \quad & \sum_{i=1}^n \kappa_i (h_i - h_i^{curr})^2 \\ \text{s.t.} \quad & Sh = s^{des} \\ & \mathbf{1}^T h + \sum_{i=1}^n \max\{-v_i h_i, 0\} \leq \mathbf{1}^T h^{curr} \end{aligned}$$

Exercise: transform this problem to a constrained linear least squares problem., and formulate its KKT conditions.